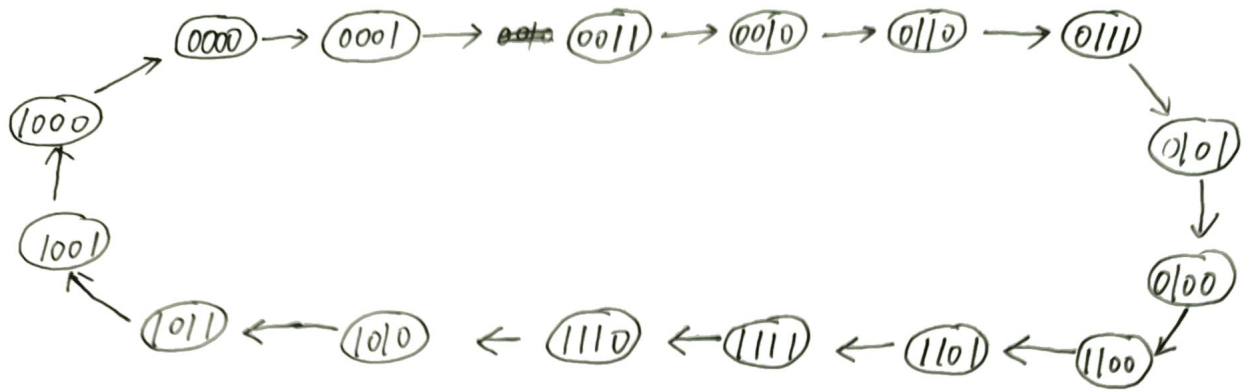




| Present State |       |       |       | Next State |       |       |       |
|---------------|-------|-------|-------|------------|-------|-------|-------|
| $C_1$         | $C_2$ | $C_3$ | $C_4$ | $N_1$      | $N_2$ | $N_3$ | $N_4$ |
| 0             | 0     | 0     | 0     | 0          | 0     | 0     | 1     |
| 0             | 0     | 0     | 1     | 0          | 0     | 1     | 1     |
| 0             | 0     | 1     | 0     | 0          | 1     | 1     | 0     |
| 0             | 0     | 1     | 1     | 0          | 0     | 1     | 0     |
| 0             | 1     | 0     | 0     | 1          | 1     | 0     | 0     |
| 0             | 1     | 0     | 1     | 0          | 1     | 0     | 0     |
| 0             | 1     | 1     | 0     | 0          | 1     | 1     | 1     |
| 0             | 1     | 1     | 1     | 0          | 1     | 0     | 1     |
| 1             | 0     | 0     | 0     | 0          | 0     | 0     | 0     |
| 1             | 0     | 0     | 1     | 1          | 0     | 0     | 0     |
| 1             | 0     | 1     | 0     | 1          | 0     | 1     | 1     |
| 1             | 0     | 1     | 1     | 1          | 0     | 0     | 1     |
| 1             | 1     | 0     | 0     | 1          | 1     | 1     | 1     |
| 1             | 1     | 0     | 1     | 1          | 1     | 1     | 0     |
| 1             | 1     | 1     | 0     | 1          | 0     | 1     | 0     |
| 1             | 1     | 1     | 1     | 1          | 1     | 1     | 0     |

卡诺图“0”化简

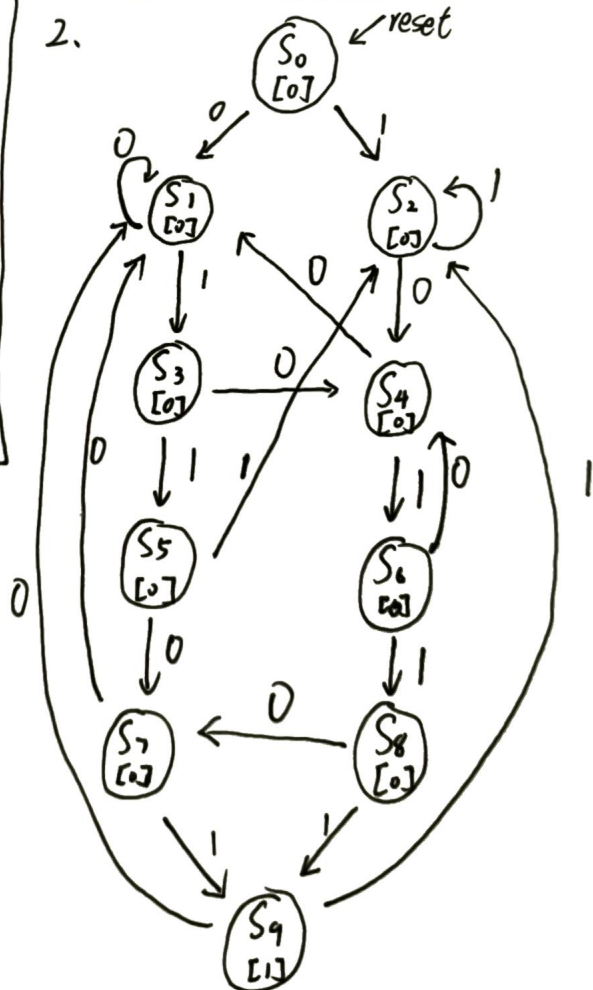
$$N_1 \leftarrow (C_2 + C_3 + C_4)(C_1 + C_4')(C_1 + C_2')$$

$$N_2 \leftarrow (C_1' + C_3' + C_4)(C_2 + C_3)(C_2 + C_4')$$

$$N_3 \leftarrow (C_3 + C_4)(C_1 + C_2' + C_4')(C_1' + C_2 + C_4')$$

$$N_4 \leftarrow (C_1 + C_2 + C_3')(C_1 + C_2' + C_3)(C_1' + C_2' + C_3')$$

2.



3.  $m$  为输入序列 0 的个数,  $n$  为 1 的个数

~~10~~

此列  $m=0$       此列  $m \bmod 3 = 1$       此列  $m \bmod 3 = 2$       此列  $m > 0$  且  $m \bmod 3 = 0$

此行  $n=0$

此行  $n \bmod 2 = 1$

此行  $n \bmod 2 = 0$  且  $n > 0$

